Abhishek Kumar Singh

Embedded Systems | Edge AI | IoT

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PROFILE

B.Tech ECE student specialising in IoT, building C++ inference runtimes, embedded sensor systems and reproducible validation tools. Seeking embedded systems, Edge AI and IoT internship opportunities.

EDUCATION

Madan Mohan Malaviya University of Technology, Gorakhpur

B.Tech, Electronics & Communication Engineering (IoT specialisation) | 2025-2029, in progress

SELECTED PROJECTS

AeroNode | C++17, PyTorch, ONNX Runtime, ZeroMQ [Source]

- Built staged inference with persistent C++ sessions, explicit tensor buffers and entropy-based early exits; verified downstream stage skipping and numerical parity.
- Frozen-policy evaluation measured **2.1052 ms** mean latency versus **2.8139 ms** full-depth, with 72.6857% versus 72.9400% accuracy on 1,966 held-out images. Scope: Windows CPU, batch 1, one thread, three repetitions.

CausalBench | Python, SQLite, JavaScript, optional C++ controllers [Source]

- Built a local lab to generate/import timestamped sensor traces, inject seeded faults, replay baseline and guarded decisions, and inspect disagreements with simulated actuation.
- Added failure minimization, saved regressions and reproducible source/evidence ZIP exports. Repository documents local validation with **93 Python tests and 13 browser workflows**; hardware is optional.

JOCKY | Rust, FastAPI, React/TypeScript, SQLite [Source]

- Developed a Rust compiler and deterministic investigation plans, read-oriented collectors, Sigma/YARA-X analysis, evidence timelines and relationship graphs.
- Implemented Ed25519 evidence checkpoints and HTML/JSON reporting. **Working local platform**; the public website demonstrates a labelled synthetic investigation.

SafeTrack (TrackSure) | ESP32-S3, FastAPI, PostgreSQL [Source]

- Built an IoT shipment-monitoring prototype using DHT11, MPU6050 and light sensing, with ATECC508A-signed events and a hash-linked ledger.
- Connected firmware to a verifying Python gateway, durable SQLite outbox and Admin/User dashboards. Scope remains a prototype; endurance testing and independent backend signature auditing remain further work.

TECHNICAL SKILLS

Languages: C, C++17, Python, Rust, TypeScript/JavaScript

Embedded & inference: ESP32-S3, I2C, MPU6050, ATECC508A, PyTorch, ONNX Runtime, CMake, ZeroMQ

Backend & tools: FastAPI, PostgreSQL, SQLite, React, Git/GitHub, Sigma, YARA-X